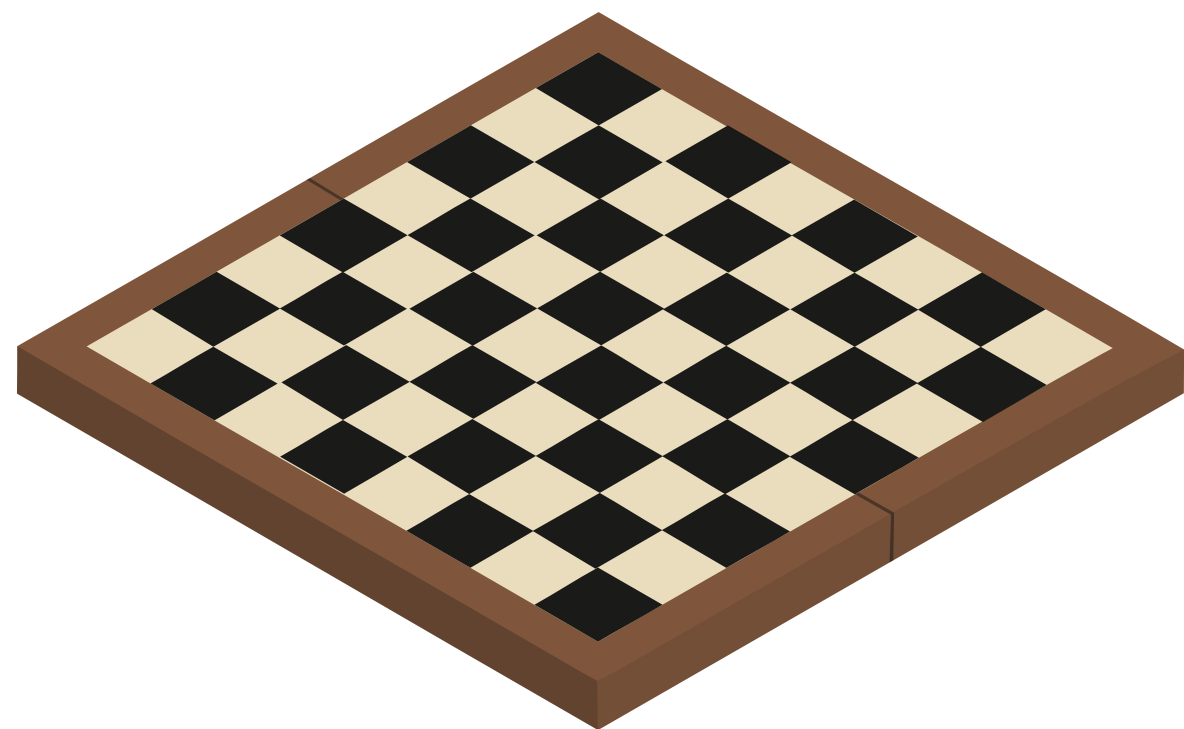
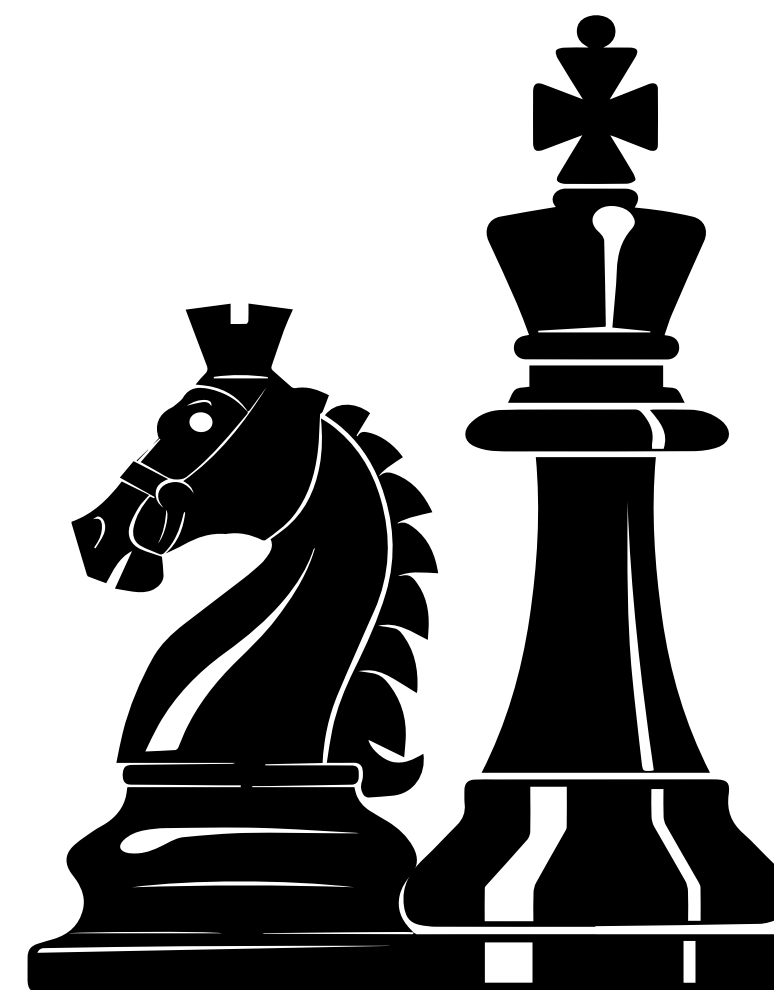




ChessPal

Chelsea Ongjoco
Ishmael Kwayisi



Overview

Our project is a web application that allows users to play chess while receiving **real-time tutoring and guidance.**

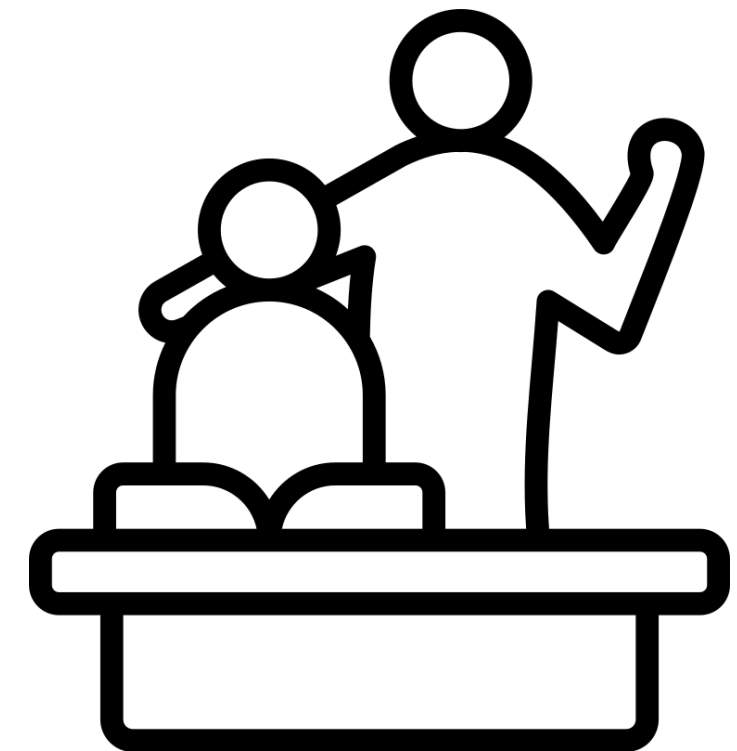
Why are we doing this?

Most chess learning tools today rely on static lessons, videos, or puzzles. While these are helpful, they do not teach players during actual gameplay, **when learning is most helpful.**

Our goal is to create an application that bridges this gap by embedding tutoring directly into a live chess match.



Only has post game analysis



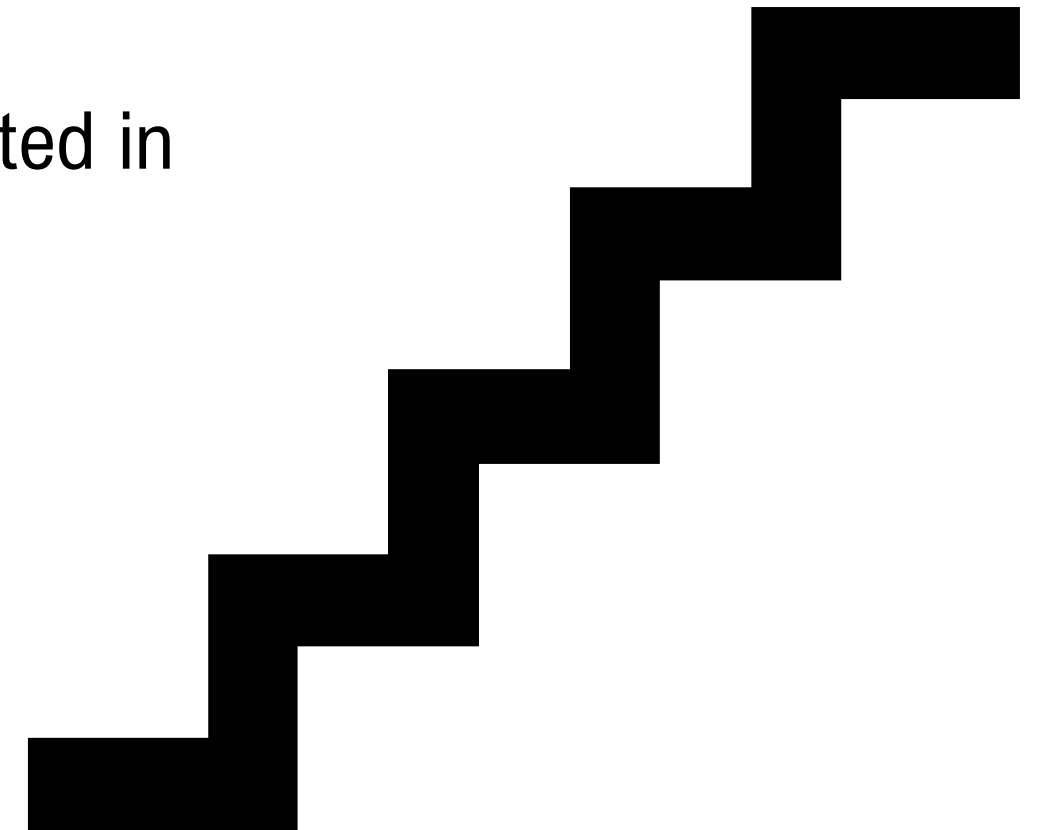
Who will benefit?

This project aims to help:

- Beginners of all ages learning chess for the first time
- Intermediate players wanting to improve strategy
- Students participating in chess clubs or competitions
- Anyone who learns best through interactive, real-time feedback

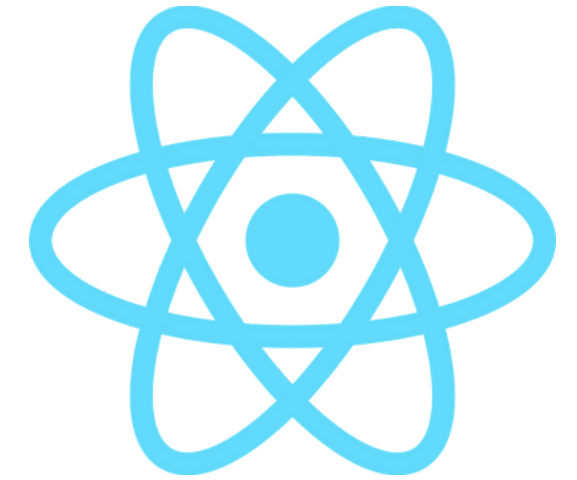
Overall, the app is designed to be accessible and valuable to anyone interested in improving their chess skills.

It is especially helpful to people with no connections to other good chess players who can correct them on the spot.



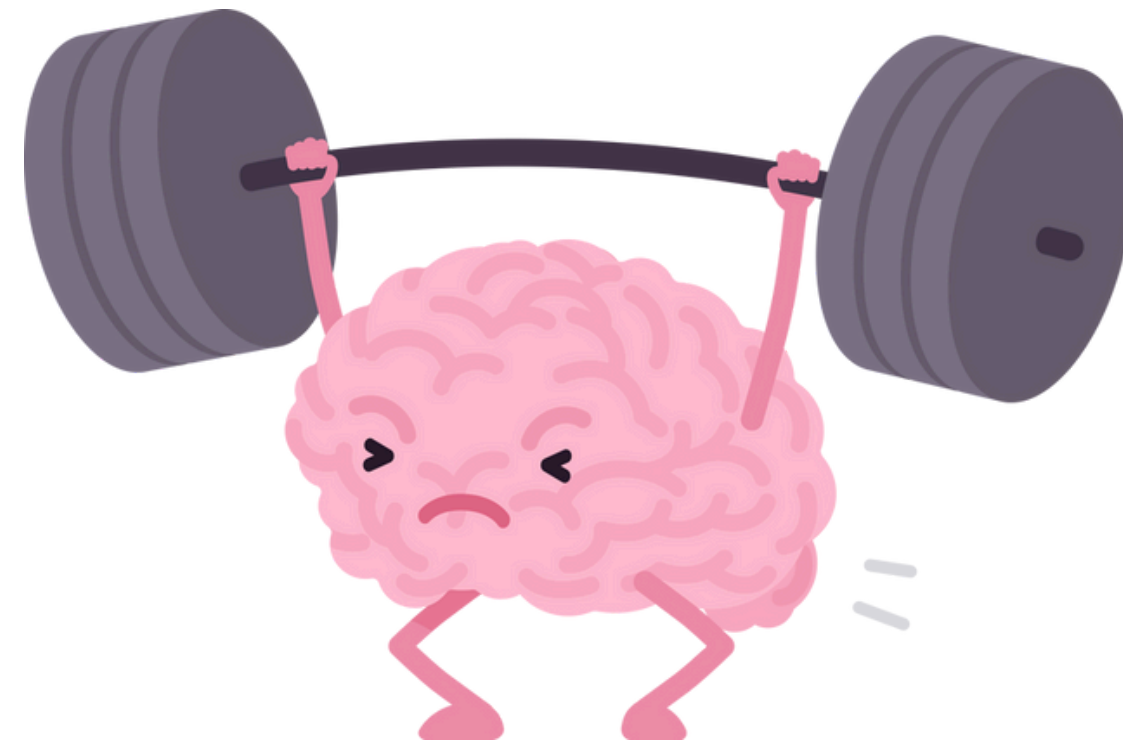
Tech Stack

- **Vite + React** (Frontend)
 - Responsive single-page app
- **Node.js** (Backend)
 - Handle routes & Server logic
- Regular **CSS**
- **Stockfish**
 - Analyzes the position and best moves based on established Chess rules and deep search analysis
 - Parses FEN strings to exchange positions reliably
- **Chat GPT API**
 - Turns the stockfish output into a more human readable output for the embedded tutor
 - Model: GPT 4o-mini (fast and cheap)



Problems we overcame

- **Chessboard learning curve** impacted development
- **Tailwind issue**
- Switching from just **ChatGPT** → **Stockfish + ChatGPT hybrid**
 - **GPT** is amazing at explanation, but terrible at tactical calculation.
 - **Stockfish** is amazing at tactics, but terrible at human explanation.
- **Unstable Stockfish**
- [Object Object] **AI output error**
- **AI Opponent** Infinite Play



In Depth slide (only if you are curious)

- **React Chessboard:** Originally, we wanted to code the chessboard ourselves, but we realized the heart and innovation of this project was applying the AI tutor to it. With this... we decided to just use pre-built code to code the chessboard so we can focus on making the AI tutor good.
- **Tailwind issues:** That code used tailwind CSS for the board and our project was not configured for that... so we had to remove the code with tailwind and just use a separate CSS page for the styling.
- **Switching from just ChatGPT to Stockfish + ChatGPT hybrid:** At first, we assumed ChatGPT alone could understand the board state and give good enough chess insights. We realized we were wrong (GPT alone was very inaccurate and hallucinates often) and we needed to use an established tool to do most of the heavy lifting... so we used **Stockfish**
 - **GPT** is amazing at explanation, but terrible at tactical calculation.
 - **Stockfish** is amazing at tactics, but terrible at human explanation.
- **Unstable Stockfish:** During development, we had trouble getting Stockfish files to be read by the program. Resulting in several crashes with our server, since it was reliant on Stockfish. To remedy this, we tried several combinations of older versions of Stockfish and Node.js, until settling on Node.js v20 and Stockfish v17.
 - Previously used Node v24 → Not an LTS release
 - Downloaded Stockfish v17.1 → Provided access to bypass WASM loaders
- **AI Opponent Infinite Play:** When AI opponent was implemented into the game, it didn't know how to stop playing its turn. The root cause for this was that there was a hard coded value included in our boardToFEN function's game state parameter. This needed to be changed to a gameState variable with a state called "turn" to let the function know that turn is dynamic.

Prompt beginning & end

Improved prompt to be more tactical and clear

Paymvi committed 3 weeks ago

Fixed SAN conversion

Paymvi committed 3 weeks ago

Adjusted Stockfish+GPT hint engine with SAN, FEN, and depth 20 search

Paymvi committed 3 weeks ago

Added Stockfish to gain better tactical insight

Paymvi committed 3 weeks ago

Improved AI prompt and styling

Paymvi committed 3 weeks ago

Added server and base AI

Paymvi committed 3 weeks ago

```
+ function buildPrompt(gameState) {
+   return `
+ You are a chess coach (2000+ ELO). Give a clear, concise, helpful hint.
+
+ Game Info:
+ - Turn: ${gameState.turn}
+ - Check: ${gameState.check}
+ - Checkmate: ${gameState.checkmate}
+ - Move history: ${gameState.moveHistory.map(m => mnotation).join(", ")}
+
+ Return JSON:
+ {
+   "suggestion": "...",
+   "explanation": "..."
+ }
+ `;
+ }
```

+ Never contradict a forcing line, capture, tactic, or hanging piece detection.

+ Never offer "strategic" ideas when a forcing tactic is available.

+ -----

+ POSITION INFO

+ -----

+ FEN: \${fen}

+ Engine best move (SAN): **`\${sanMove}`**

+ Player's last move: **`\${lastMoveSAN}`**

+ Move history: \${gameState.moveHistory.map(m => mnotation).join(", ")}
+
+ -----
+ WHAT YOU MUST DO
+ -----
+
+ ### 1. Evaluate the player's last move
+ - Use one of these labels: **Blunder**, **Mistake**, **Inaccuracy**, **Good**, **Excellent**
+ + Always give a **tactical** reason (e.g. "missed pawn capture", "left a piece hanging", "ignored forcing move").
+ - Use at least 1 emoji.
+ - Keep it factual, concise, non-fluffy.
+
+ ### 2. Recommend the engine move (**`\${sanMove}`**)
+ - State clearly **why** the engine prefers it.
+ - Emphasize the tactical reason: winning material, preventing loss, gaining tempo, forcing sequence, exploiting a pin, etc.
+ - Keep this section **tight and logical**.
+
+ ### 3. Explain the position (8-12 sentences MAX)
+ - Explain key tactical motifs only (pins, forks, discovered attacks, undefended pieces, forcing captures).
+ - Include **only** the relevant positional ideas (development, center control) but **no long-term speculative plans**.
+ - Use **clean, simple language**, minimal fluff.
+ - Include **1-3 emojis** (no more).
+
+ ### HARD RULES
+ - ❌ No long variations
+ - ❌ No engine-style evaluation numbers
+ - ❌ No fictional "later ideas" unless directly tied to tactics
+ - ❌ No soft storytelling or filler
+ - ✅ Short, direct, tactical, factual
+ - ✅ JSON output only

 stockfish!!!

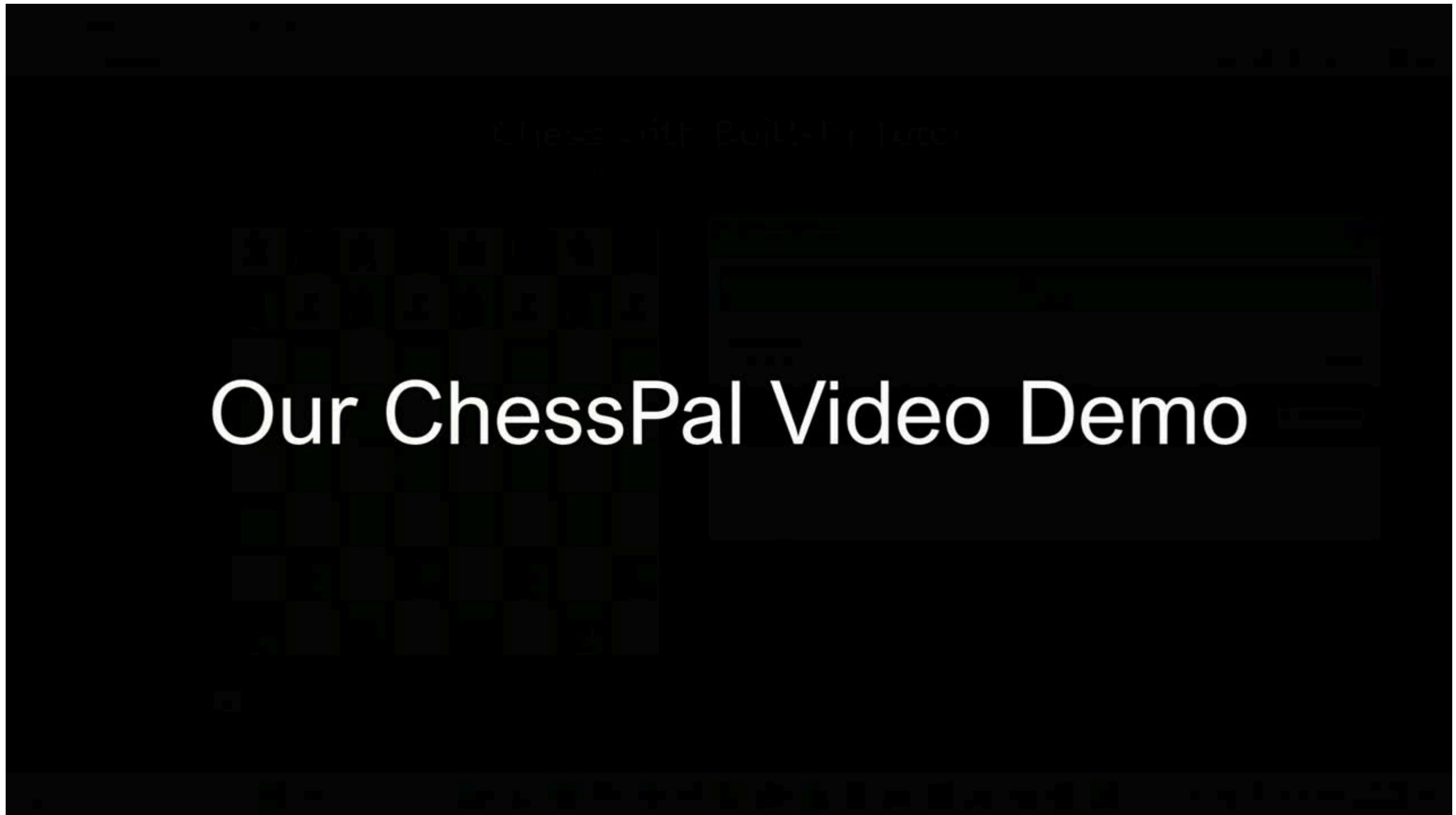
Team contributions

We had really good communication throughout the entire project!
This helped us split the work evenly while staying flexible.

- Chelsea added in the chessboard code
- Ishmael imported tailwind
- Chelsea changed tailwind to regular CSS
- Chelsea added in the original AI (just ChatGPT)
- Chelsea transitioned the AI to use Stockfish (with a more tactical prompt)
- Ishmael fixed the Stockfish issues (while keep prompt integrity)
- Ishmael added a voice feature & adjusted the hint button to only be active during valid times
- Ishmael added opponent AI
- Chelsea restructured the file structure to separate the client and server content
- Chelsea fixed the [object object] problem in the AI output
- We both worked on the documentation (including the .md files)



Our ChessPal Video Demo



Testing the AI

To reduce hallucinations, we told GPT to give Stockfish the utmost authority. This reduced a lot of hallucinations, but they still do exist unfortunately.

When it comes to the **foundational properties** like getting the chessboard implemented, moves being valid, the AI opponent (fully stockfish) working properly, all of our tests passed for those.

[Link to tests](#)

However, we noticed that our AI (Stockfish + GPT) tutor **still isn't perfect** even after all the iterations... for example:

- Sometimes it comments on black's move
- Sometimes the logic is correct but the output like "c4" is wrong and vice versa
- Sometimes it thinks ahead too much

Learn and improve your chess skills with real-time guidance

Chess Tutor

Hint It means to write "c3" not c4
Best move is c4 to immediately challenge the knight on b4 and gain more space. 🚀

Evaluation of Last Move Talks about black's knight
Mistake: The move Nxb4 overlooks the tactical opportunity to solidify White's position. While it captures a pawn, it neglects the potential to enhance piece activity and opens up the position for White. ⚠️

Position Summary It assumes you will take its advice
The current position sees Black's knight on b4 being threatened by White's pawn on c4. This gives White an opportunity to gain tempo and increase central control. White currently has a slight space advantage, with pawns further advanced. Black's pieces are somewhat passive, and the knight's future is now in jeopardy. The king's safety is adequate for both sides at this stage. If c4 is played, White will dictate the flow of the game by forcing the knight to move. 🏹 ✨

Tactical Motifs

- Knight under threat at b4
- Central pawn control
- Pawn structure advantage
- Potential for piece activity increase
- Tactical pressure on e5

Get Hint

Moves played:
Current turn: 2 White

Bigger additions for the future...

Realistic additions:

- Have different buttons for different AI difficulties
- Have a login which stores more personal information
- Add in options to customize the look of the chessboard
- Making the AI load faster/more responsive

“Stretch” goals:

- Be able to battle others online (still with the AI insights)
- Be able to ask the AI questions mid game
- It would be cool to have a mobile app version of this





Questions?

Thank you!